Fast Minds, Slow Light

Causal Limits on the Growth of Computational Civilizations

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Abstract

The computational inward transition hypothesis (CITH) treats the delay of remote returns as a parameter. The speed of light makes that delay geometric and constrains growth in two further ways. We derive the consequences within the resource-allocation model of CITH. First, causality bounds acquired resources by the content of the light cone: with bounded resource density the budget grows at most polynomially, any exponential acquisition phase has a computable maximum duration, and in flat ΛCDM nothing emitted from now on can influence matter beyond a comoving radius of 16.7 billion light-years. A causal cap on the acquisition process of CITH weakly enlarges its inward basin but cannot undo a crossing of the tipping budget. Second, for a subjective clock ω and discount rate δ, the latency length $\ell_s = c/(\delta\omega)$ converts the robustness ratio ρ into a distance: the corner test for the nearest opportunity holds exactly when that opportunity lies beyond $(\ell_s/\kappa)\ln\rho$. Centrally evaluated frontier expansion therefore stops at a finite radius, asymptotically $(k/\kappa)\ell_s \ln \ell_s$ for resource content growing as r^k, and geographical backups lag by $\omega r/c$ in subjective time. Third, integration within a subjective round trip τ confines computation to a diameter $c\tau/(2\omega)$, so integrated power has a ceiling falling as ω^{-2}. Combining these channels yields a unique latency-containment budget, from which c cancels at the fastest integration-compatible clock. None of these limits binds autonomous descendants, whose lineages grow as percolation clusters. The speed of light thus supplies physical reasons for the bounded budgets on which inward scenarios rely, but not, by itself, an explanation of silence.

Keywords: Fermi paradox; speed of light; latency; computational civilizations; resource acquisition; percolation; technosignatures.

1 Introduction

The computational inward transition hypothesis asks why a civilization able to expand would stop paying to do so (Joo, 2026). Its eighth version (hereafter v8) answers with a resource-allocation theory: a corner criterion, a tipping budget, an inward basin, a no-replication criterion, and a population constraint on the expansionist tail. Distance appears in v8 only through a parameter. A remote project has an objective delay T^{lag}, a civilization has a subjective clock ω, and the discount exponent is $\delta \int \omega \, \mathrm{d}u$ (v8, eq. (58) and Proposition 24). Its main illustration of delay is a four-light-year trip at a tenth of the speed of light.

The speed of light is not one delay among many. It fixes how fast physical resources can be acquired, the minimum delay of every remote return, and the largest region that can compute as a unit. Each of these limits interacts with the clock. For a mind running ω times faster than physical time, light covers c/ω per subjective second: about $300\,\mathrm{m\,s^{-1}}$ at $\omega = 10^6$. At that clock one astronomical unit is 15.8 subjective years away, and the nearest known stellar system, at about 1.30 pc (Reylé et al., 2021), is more than four million subjective years away. Sandberg

(1999) noted that faster processing makes the internal light delays of large computing objects subjectively longer, and Hanson (2016) discussed how communication delays would limit the separation of fast emulated minds in dense cities. This paper turns those observations into allocation results connected to v8.

1.1 Contributions

Each contribution below is conditional on stated assumptions and is proved in appendix A; numerical checks are listed in appendix B.

1. *Acquisition* (section 3). Causality bounds acquired resources by the content of the light cone (lemma 3). With bounded density the budget is at most polynomial in time, and an exponential phase has an explicit maximum duration (corollary 4). A causal cap on v8's acquisition process weakly enlarges the inward basin but leaves lock-in intact (proposition 5). In flat ΛCDM the affectable budget is finite (lemma 6).
2. *Latency* (section 4). Corner tests in replication directions alone certify no replication (lemma 7). The latency length turns v8's robustness ratio into a minimum distance to the nearest opportunity (proposition 8). Centrally evaluated frontier expansion stops at a finite radius scaling as $\ell_s \ln \ell_s$ (proposition 9). Geographical backups are stale by $\omega r/c$ in subjective time (corollary 10).
3. *Integration* (section 5). A computation that must complete subjective round trips within τ has a power ceiling falling as ω^{-2} (proposition 13). We resolve the coupling of computational efficiency and clock speed that v8 left as future work (proposition 15), and obtain a latency-containment budget from which c cancels (theorem 16).
4. *Decentralization* (section 6). Autonomous colonies face nearest-neighbour delays, not the distance to their origin. Their lineages are percolation clusters, so restraint must be inherited colony by colony (proposition 17). Near-light fronts also hide themselves until they arrive (remark 18).

1.2 Relation to prior work

Physical limits on computation are established subjects. Lloyd (2000) bounded the operation rate and memory of a computer by its energy and degrees of freedom, using the bound of Margolus and Levitin (1998); Sandberg (1999) reviewed density, speed, dissipation and communication limits on very large computing objects; and Bradbury (1999) described stellar-scale nested computers. Cosmological acceleration limits what any civilization can reach or compute (Krauss and Starkman, 2000; Armstrong and Sandberg, 2013; Ord, 2021). Expansion fronts and their visibility have been modelled as bubbles in an expanding universe (Olson, 2015; Hanson et al., 2021); settlement fronts and their saturation by Newman and Sagan (1981) and Carroll-Nellenback et al. (2019); and colonization as percolation by Landis (1998). Slowing computation as the universe cools was proposed by Dyson (1979) and, as deliberate waiting, by Sandberg et al. (2017). Movement toward dense internal scales was described by Smart (2012). The present paper claims no priority for these qualitative ideas. Its contribution is the set of inequalities that connect them to the corner, basin, no-replication and population results of v8.

1.3 Scope

All results are conditional. We do not estimate subjective clocks, discount rates, integration tolerances or heat-flux limits for any civilization, and the examples use scenario values labelled as such. Space is static and Euclidean except in section 3.3. Clocks are constant unless stated. For a non-decreasing clock, v8's integral exponent is at least the value computed with the clock at the decision time, so every sufficient condition below remains sufficient.

2 Causal geometry and subjective scales

2.1 Inherited objects

We use v8's notation without change. A civilization allocates a budget B between local investment, with production $a\ell(z)$, and external directions j with prices p_j, initial computational returns r_j and residual slopes S_j. Value is F_η with curvature η; $d_0 = 1 + C_0$, $\Omega = a\ell(B)$ and $\Gamma = \ell(B)/\ell'(B)$. The all-inward corner is optimal under concavity exactly when $p_j \geq P_j$ in every direction, where

$$P_j = \frac{r_j + S_j(d_0 + \Omega)^\eta}{a\ell'(B)} \tag{1}$$

is the *required price* of v8, Proposition 3. For $\eta = 1$ and $\ell(z) = \ln(1 + z)$, $P_j = H_j + A_j/a$ with $H_j = (1 + B)\ln(1 + B)S_j$ and $A_j = (1 + B)(r_j + d_0 S_j)$, and v8's robustness ratio is $\rho = \max_j P_j/p_j$. The tipping budget B_{tip} solves $\Gamma(B_{\text{tip}}) = \min_{S_j>0} p_j/S_j$ (v8, Proposition 11).

2.2 Three lengths

A project in direction j at distance r has an objective delay

$$T_j(r) \geq T_j^0 + \kappa_j \frac{r}{c}, \tag{2}$$

where $T_j^0 \geq 0$ collects construction and deployment time. An effect that must travel out at speed $v \leq c$ has $\kappa_j \geq c/v \geq 1$. A payoff that must also be reported home has $\kappa_j \geq c/v + 1 \geq 2$. For a constant clock the discount exponent of v8 is $D_j(r) = \delta\omega T_j(r)$.

Definition 1 (Subjective scales)**.** For a constant clock $\omega > 0$, a subjective discount rate $\delta > 0$ and a subjective integration tolerance $\tau > 0$, define the *subjective light speed* c/ω, the *latency length* and the *coherence diameter*

$$\ell_s = \frac{c}{\delta\omega}, \qquad \Lambda = \frac{c\tau}{2\omega}. \tag{3}$$

The latency length is the distance over which a one-way light delay adds one unit to the discount exponent. The coherence diameter is the largest separation across which a signal round trip fits within τ subjective time.

Lemma 2 (Scale identity)**.** *The ratio $\ell_s/\Lambda = 2/(\delta\tau)$ depends only on subjective quantities. For a computation of diameter Λ, a round trip $(\kappa = 2, T^0 = 0)$ to distance r has discount exponent*

$$D = \delta\tau \, \frac{r}{\Lambda}. \tag{4}$$

The proof is division of the two expressions in equation (3). The identity explains why the speed of light will cancel from the containment result of section 5. Both the size of an integrated computation and the distance at which it discounts remote returns are proportional to c/ω.

Table 1 lists the three lengths at three clocks. At $\omega = 10^6$, a subjective day of round-trip tolerance spans about one Earth diameter, and a subjective year spans 0.032 astronomical units. Figure 1(a) plots Λ against ω. A kilo-speed clock with a tolerance of 0.1 s, a reaction time, gives 15 km. This is the scale of city-size separation discussed by Hanson (2016).

3 Acquisition: the light-cone budget

3.1 The causal envelope

Place the origin at the centre of a home ball of radius r_h. Let $M(r)$ be the resource content, in v8's budget units, of sites within distance r of the origin but outside the home ball. Thus

Table 1: Subjective scales for three clocks. One-way light times are in subjective units. Distances use the IAU astronomical unit and parsec and the JPL mean Earth radius (JPL, 2026); 1.30 pc is the distance of the nearest known stellar system (Reylé et al., 2021). Clock speeds and tolerances are scenarios.

	$\omega = 1$	$\omega = 10^3$	$\omega = 10^6$
Subjective light speed c/ω ($\mathrm{m\,s^{-1}}$)	2.998×10^8	2.998×10^5	299.8
Earth diameter, one way	0.0425 s	42.5 s	11.8 h
1 au, one way	8.32 min	5.78 d	15.8 yr
1.30 pc, one way	4.24 yr	4,240 yr	4.24×10^6 yr
Λ at $\tau = 0.1$ s	1.50×10^7 m	15.0 km	15.0 m
Λ at $\tau = 1$ day	86.6 au	0.0866 au	1.30×10^7 m
Λ at $\tau = 1$ yr	0.153 pc	31.6 au	0.0316 au

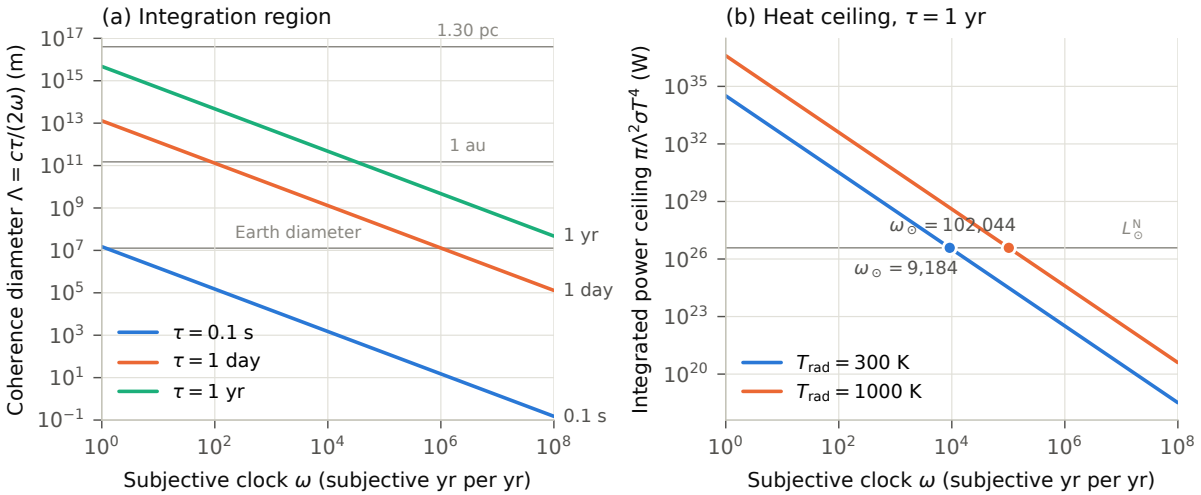


Figure 1: (a) Coherence diameter $\Lambda = c\tau/(2\omega)$ for three subjective tolerances, with reference lengths. (b) The integrated power ceiling $\pi\Lambda^2\sigma T_{\mathrm{rad}}^4$ of proposition 13 for $\tau = 1$ yr and two radiator temperatures. Dots mark the clock $\omega_\odot$ above which the nominal solar luminosity can no longer be dissipated by one integrated computation. All tolerances, temperatures and clocks are scenarios.

M is non-decreasing and vanishes on $[0, r_h]$. A uniform density n of sites with mean content $\bar{b}$ gives $M(r) \approx (4\pi/3)n\bar{b}\, r^3$. Roughly 340 stellar and substellar systems lie within 10 pc of the Sun (Reylé et al., 2021), a mean system density of about $0.08\,\mathrm{pc}^{-3}$. In a disc much wider than it is thick the growth is quadratic, and beyond a galaxy M has long plateaus.

Lemma 3 (Causal acquisition envelope)**.** *Let an acquisition process be initiated in the home ball at objective time zero, with no influence propagating faster than c. By time $t \geq 0$, the resources at sites it has influenced are at most $M(r_h + ct)$, and the resources it has delivered back into the home ball are at most $M(r_h + ct/2)$. If $M(r) \leq Kr^k$ for $r \geq r_h$, then a budget $B(t)$ built from the home content B_{home} and influenced sites satisfies*

$$B(t) \leq B_{\mathrm{home}} + K(r_h + ct)^k, \qquad \limsup_{t\to\infty} \frac{\ln B(t)}{t} \leq 0. \tag{5}$$

Influence must lie in the causal future of the home ball, and a delivery requires a round trip. The proof is in appendix A.1.

Corollary 4 (Finite exponential phase)**.** *Under the power envelope, if $B(t) \geq B(0)\exp(gt)$ on $[0, T]$ with $g > 0$ and $B(0) > 0$, then*

$$T \leq \max\left\{ \frac{r_h}{c},\ \frac{(B_{\mathrm{home}}/K)^{1/k}}{2c},\ 2c\left(\frac{2K}{B(0)}\right)^{1/k}\left(\frac{2k}{eg}\right)^2 \right\}. \tag{6}$$

The bound uses $e^y \geq (ey/m)^m$ with $m = 2k$; see appendix A.1. It applies directly to v8's inward-basin model. In v8, Proposition 14, a path that reaches the tipping budget satisfies $B_{t+1} - B_{\mathrm{tip}} > (1 + g_X/p)(B_t - B_{\mathrm{tip}})$ (v8, eq. (43)), so its excess grows at least geometrically. With any physical period length and any polynomial envelope, that runaway is causally consistent only for finitely many periods. The model needs a cap.

3.2 A causal cap on the inward basin

Retain the setting of v8, Proposition 14: one replicating channel, $p, S > 0$, $r \geq 0$, $d_0 \geq 1$, $X_0 = 0$, a mobilized local budget $0 < \bar{B} < B_{\mathrm{tip}}$, productivity $a_t = a_0 \exp(g_a t)$, and the myopic static policy $x_t = x^*(B_t, a_t)$ of v8, eq. (42). Replace the acquisition rule by

$$B_t = \bar{B} + \min\{g_X X_t, E_t\}, \qquad X_{t+1} = X_t + x_t, \tag{7}$$

where $E_t \geq 0$ is non-decreasing. For example, $E_t = M(r_h + c\Delta t/2)$ for a period of length Δ. Investment beyond the cap is banked as stock in transit and becomes usable as the cap rises.

Proposition 5 (A causal cap enlarges the inward basin)**.** *In the capped model:*

- *(i) every budget path lies below the uncapped path from the same a_0, is non-decreasing in t, and is pointwise non-increasing in a_0;*
- *(ii) if the uncapped path from a_0 is captured, so is the capped path. The captured initial productivities form an up-set containing $(a_{\mathrm{sep}}, \infty)$, so its infimum, the capped separator, satisfies $a_{\mathrm{sep}}^{\mathrm{cap}} \leq a_{\mathrm{sep}}$;*
- *(iii) no capped path reaches B_{tip} before $t_E = \min\{t : \bar{B} + E_t \geq B_{\mathrm{tip}}\}$, and if $\bar{B} + \sup_t E_t < B_{\mathrm{tip}}$ every positive a_0 is captured;*
- *(iv) once a capped path reaches B_{tip}, new external investment is positive in every later period, while $B_t \leq \bar{B} + E_t$.*

The proof inducts on t, using v8's monotonicity of x^* in budget and productivity (appendix A.2). Light speed thus delays lock-in and caps its growth rate, but cannot reverse a crossing.

Table 2 applies the cap to v8's numerical example. Slower envelopes lower the separator from 6.632 to about 4.33, but not to zero. While the cap binds, a low-productivity path keeps investing for the residual payoff Sx. The banked stock becomes usable as the light cone grows, and the path then crosses the tipping budget. Only a bounded envelope, as in (iii), captures every path.

Table 2: v8's example $(p, S, r, d_0) = (1, 0.002, 0.05, 1)$, $g_a = 0.05$, $g_X = 0.5$, $\bar{B} = 0.6B_{\text{tip}}$, with the cubic cap $E_t = e_3 t^3$. Here t_E is the first period at which the cap could admit B_{tip}. The uncapped separator is 6.632159. Separators are bisected on certified capture or crossing.

e_3	t_E	$a_{\text{sep}}^{\text{cap}}$
1	3.49	6.612150
10^{-1}	7.51	6.454066
10^{-2}	16.19	5.882275
10^{-3}	34.87	4.950497
10^{-4}	75.13	4.404970
10^{-5}	161.85	4.334291

3.3 A cosmological ceiling

Lemma 6 (Finite affectable budget)**.** *In a flat Friedmann–Lemaître–Robertson–Walker universe with $H(a) = H_0 E(a)$, $E(a) = [\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda]^{1/2}$ and $\Omega_\Lambda > 0$, a signal emitted at scale factor a_e reaches comoving distance at most*

$$\chi_{\text{EH}}(a_e) = \frac{c}{H_0} \int_{a_e}^{\infty} \frac{\mathrm{d}a}{a^2 E(a)} \leq \frac{c}{H_0 a_e \sqrt{\Omega_\Lambda}}, \tag{8}$$

and material probes reach strictly less. If usable resources have bounded comoving density $\bar{n}$, every budget built from sites influenced after today is at most $B_{\text{home}} + (4\pi/3)\bar{n}\chi_{\text{EH}}(1)^3$.

For the Planck 2018 base-ΛCDM values $H_0 = 67.4\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ and $\Omega_m = 0.315$ (Planck Collaboration, 2020), flat and without radiation, $\chi_{\text{EH}}(1) = 16.68$ billion light-years (5.11 Gpc), against the bound 17.53. The particle horizon is 47.0 billion light-years (46.1 with a radiation density of 9.2×10^{-5} of critical). The affectable comoving volume is therefore 4.5–4.8 per cent of the currently observable volume. These are kinematic statements, consistent with Krauss and Starkman (2000) and Ord (2021).

The consequence for CITH is logical rather than practical. v8's unbounded-budget theorems (Proposition 8 and Corollaries 9 and 19) describe intermediate regimes. The final classification compares a bounded budget with the tipping budget (v8, Proposition 11) or with its ceiling version (Corollary 12). Since $\Gamma(B) \geq B$, the tipping budget is at most $\min_j p_j/S_j$. A cosmic budget therefore helps all-inward allocation only if residual slopes are smaller than p_j/B_{reach}, which is extraordinarily small.

4 Latency: remote returns on a subjective clock

4.1 Replication tests certify no replication

The latency results concern replication. Observation and dormant backups may remain active at short range, so we first separate the two.

Lemma 7 (Corner tests certify no replication)**.** *Take v8's model with observation and backups (v8, eq. (51) and Proposition 20): concave ℓ and residual functions, and strictly increasing concave F. Suppose a no-replication face optimizer exists with finite derivatives. If each replication direction $j \in \{n, h\}$ passes its all-inward corner test,*

$$F'(C_L) r_j + S_j \leq p_j F'(C_L) a\ell'(B), \qquad C_L = C_0 + a\ell(B), \tag{9}$$

then a no-replication optimizer is globally optimal. Strict inequalities exclude replication from every full optimizer. This holds whatever the observation and backup prices, and whether or not those channels pass their own corner tests.

The face multiplier satisfies $\lambda \geq F'(C^*)a\ell'(z^*) \geq F'(C^*)a\ell'(B)$, while $F'(C^*) \geq F'(C_L)$ and $R'(b^*) \leq R'(0)$, $I'(w^*) \leq I'(0)$. Together these bounds show that equation (9) implies v8's no-replication inequalities (appendix A.4). The converse is false. In 1,200 random instances in the validator, 79 have no replication despite failing a replication corner test. On v8's frozen 1,024-row design, the 333 rows certified by equation (9) lie within v8's 421 no-replication rows, with no exception.

4.2 The latency radius

Extend v8's static model to finitely many distance classes. Direction j is available at distances r in a finite set $\mathcal{R}_j$, at price $p_j(r)$. At the all-inward corner, the right derivative of utility in that direction is $e^{-D_j(r)}[F'(C_L)r_j + S_j]$: v8's undiscounted return, attenuated by the discount exponent of the delay equation (2). Utility is concave.

Proposition 8 (Latency radius)**.** *The corner is optimal exactly when $p_j(r) \geq e^{-D_j(r)}P_j$ for every j and every $r \in \mathcal{R}_j$, with P_j from equation* (1)*. If each p_j is non-decreasing in distance, only the nearest distance $r_j^{\min} = \min \mathcal{R}_j$ matters. For a constant clock, the corner holds exactly when, in every direction with $\rho_j = P_j/p_j(r_j^{\min}) > 1$,*

$$r_j^{\min} \geq r_{\mathrm{lat},j} = \frac{\ell_s}{\kappa_j}\left[\ln \rho_j - \delta\omega T_j^0\right]. \tag{10}$$

By lemma 7, the same inequality imposed on replication directions alone certifies no replication.

This is v8, Proposition 3, applied to the enlarged set of directions (appendix A.5). The geometric content is the conversion: v8's robustness ratio becomes a distance, measured in latency lengths. Raising productivity shortens the latency radius only logarithmically.

Figure 2 applies a common delay exponent D to every external direction of v8's frozen design. This would be the case if all opportunities lay at the nearest system. The all-inward count rises from 219 at $D = 0$ to all 1,024 rows at $D = 8.539$, the largest $\ln \rho$ in the design. Rows certified by lemma 7 rise from 333 and reach 1,024 at $D = 7.950$. The exact restricted-face count rises from v8's 421 and reaches 1,024 by $D = 8$. For a constant clock, $\kappa = 2$ and a nearest system at 1.30 pc, every row is all-inward once $\ell_s \leq 0.3045$ pc, that is, once $\delta\omega \geq 1.007$ per objective year. At $\omega = 10^6$ this is $\delta \approx 1.0 \times 10^{-6}$ per subjective year, the order of the thresholds in v8's Table 1. As in v8, these counts check an implementation on a frozen synthetic design. They are not prevalence estimates.

4.3 Central frontier expansion stops

Let a civilization hold every site within a frontier radius R and evaluate all projects centrally. Its budget $B(R)$ may grow with R through imported resources. Its replication opportunities lie at distances at least R. A replication direction at distance r enters lemma 7 with computational return $e^{-D_j(r)}r_j$ and residual slope $e^{-D_j(r)}S_j$, and its marginal value at any face allocation is bounded by these discounted values. Observation and backups at held sites are left unrestricted.

Proposition 9 (Self-limiting central expansion)**.** *Assume $0 \leq \eta \leq 1$; concave ℓ with $\Gamma(B) \leq K_\Gamma(1+B)^m$ for $B \geq B_0 := B(0)$; productivity $a \geq a_{\min} > 0$; prices at least $p_{\min} > 0$; undiscounted coefficients $r_j \leq \bar{r}$ and $S_j \leq \bar{S}$, possibly varying with R; budgets $B_0 \leq B(R) \leq B_0 + K(1+R)^k$; and discount exponents $D_j(r) \geq \kappa r/\ell_s$ with $\kappa > 0$. Put $K_1 = (\bar{r} + d_0\bar{S})/(a_{\min}\ell(B_0)) + \bar{S}$ and*

$$\bar{R} = \sup\Big\{R \geq 0 : \ \frac{\kappa R}{\ell_s} \leq \ln \frac{K_1 K_\Gamma[1 + B_0 + K(1+R)^k]^m}{p_{\min}}\Big\}. \tag{11}$$

(i) $\bar{R} < \infty$. For every $R > \bar{R}$ every replication direction passes strictly, so no full optimizer at frontier R replicates.

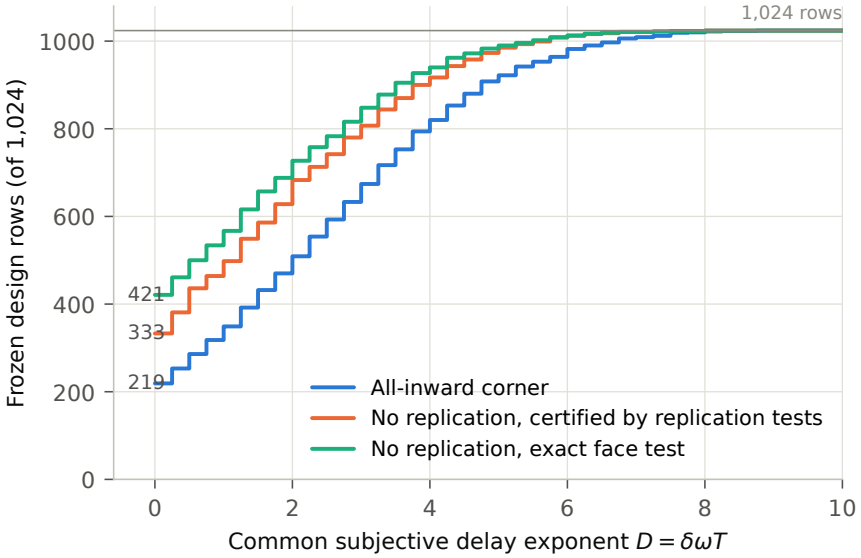


Figure 2: Rows of v8's frozen design that pass, under a common subjective delay exponent D applied to every external return and residual slope. The three tests are the all-inward corner, the replication-only certificate of lemma 7, and the exact no-replication test on the observation face. For a constant clock, $D = \kappa r_1/\ell_s$. Values at $D = 0$ reproduce v8 exactly.

(ii) $\limsup_{\ell_s\to\infty} \bar{R}/(\ell_s \ln \ell_s) \le km/\kappa$. *For logarithmic production any* $m > 1$ *is admissible, so the bound is* k/κ.

(iii) In v8's one-channel benchmark, with logarithmic production and value, constant a, p, r, S *and* $r + S > 0$, $B(R) = B_0 + KR^k$ *and* $D(R) = \kappa R/\ell_s$, *the largest radius at which expansion is optimal satisfies*

$$R_{\text{stop}} = \frac{k}{\kappa}\,\ell_s \ln \ell_s \left[1 + O\!\left(\frac{\ln\ln \ell_s}{\ln \ell_s}\right)\right]. \tag{12}$$

Delay grows linearly in the frontier radius, while the obstruction of v8 grows like $B \ln B$, only polynomially in R. Exponential discounting therefore wins (appendix A.6). In physical units, with lengths measured in a fixed unit ℓ_0 implicit in K, $R_{\text{stop}} \approx (kc/\kappa\delta\omega)\ln[c/(\delta\omega\ell_0)]$. The size of a centrally governed civilization scales with the speed of light, up to a logarithm. Letting $\ell_s \to \infty$, with instantaneous returns, recovers the growth obstruction of v8, Proposition 8.

Figure 3 uses scenario values: the v8 coefficients with $a = 5$, $B_0 = 0.6B_{\text{tip}}$, $K = 4.1$ budget units per pc^3, $k = 3$, $\kappa = 2$ and $r_1 = 1.30\,\text{pc}$. The radius reached from the home matches R_{stop} once expansion starts. It starts only for $\ell_s > 7.754\,\text{pc}$. For $4.24 \lesssim \ell_s < 7.754\,\text{pc}$, expansion would pay at radii of 10–30 pc, but the first step to 1.30 pc does not. Latency can thus stop growth at the first step even when a larger civilization would keep expanding. The ratio $R_{\text{stop}}/(\ell_s \ln \ell_s)$ converges slowly, from 2.12 at $\ell_s = 10$ to 1.69 at $\ell_s = 10^{14}$ in the same units. The scaled deviation in equation (12) stays between 1.70 and 1.72, as the proof predicts.

4.4 Stale backups

Light speed also limits geographical insurance. A backup at distance r can be updated only by signals. At a catastrophe it therefore restores the home state as of at least r/c earlier, which is $\omega r/c$ in subjective time.

Corollary 10 (Stale geographical backups)**.** *In v8's endogenous-risk model (Proposition 23), let the restored value be* $\psi F_\eta(C)$ *with* $\psi = \Psi(\omega r/c) \in [0, 1]$, *where* Ψ *is non-increasing and* $\Psi(0) = 1$.

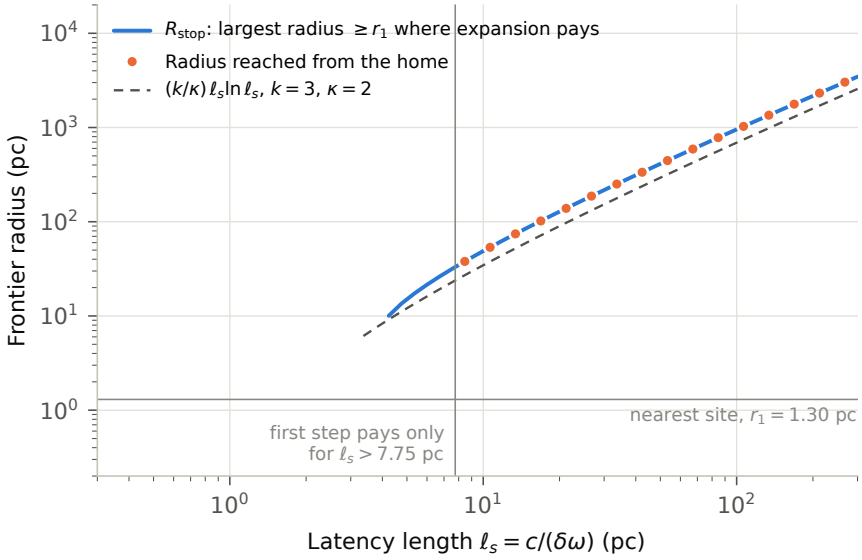


Figure 3: Central frontier expansion in the one-channel benchmark. The line is the largest radius beyond the nearest site at which expansion is optimal. Dots mark the radius actually reached from the home. The dashed curve is the leading asymptote of equation (12). All parameters are scenarios.

Then the necessary condition for no backup is v8, eq. (55), with $\varphi'(0)$ replaced by $\psi\varphi'(0)$:

$$p \geq \frac{\Gamma r}{\Omega} + \frac{\vartheta}{1-\vartheta}\psi\varphi'(0)\Gamma\Omega^{-\gamma}K_\eta(\Omega). \tag{13}$$

The limiting hazard ceiling becomes $\vartheta \leq p(1-\eta)/[p(1-\eta)+\psi\varphi'(0)\Gamma]$. If valued internal state grows at subjective rate g, then $\Psi(u) = e^{-gu}$, and staleness acts as subjective discounting at rate g over the one-way light time, even when $\delta = 0$.

At $\omega = 10^6$, a backup at 1.30 pc lags by 4.24 million subjective years. Interstellar insurance of a fast civilization therefore insures its remote past. A mirror that runs forward independently is not stale, but it diverges: it is a separate lineage, treated in section 6.

5 Integration: how large can a fast mind be?

5.1 A coherence ceiling

Definition 11 (Integration)**.** A computation occupying a bounded set $\Sigma \subset \mathbb{R}^3$ is *τ-integrated* at clock ω if every pair of its points can complete a signal round trip within subjective time τ: $2|x-y|\omega/c \leq \tau$. Then $\operatorname{diam}\Sigma \leq \Lambda$.

Signals in matter are slower than c, so the definition is optimistic. The tolerance τ is a property of the computation: a fraction of a second for one mind, perhaps days or years for a coordinated society.

Lemma 12 (Hull geometry)**.** *If $\operatorname{diam}\Sigma \leq \Lambda$, the convex hull of Σ has surface area at most $\pi\Lambda^2$, with equality for a ball of diameter Λ.*

By Cauchy's formula, the surface area of a convex body in $\mathbb{R}^3$ is four times its mean projected area (Schneider, 2014). Each projection has diameter at most Λ, hence area at most $\pi\Lambda^2/4$ by the planar isodiametric inequality (Bieberbach, 1915).

Proposition 13 (Coherence ceiling on integrated power)**.** *Suppose a τ-integrated computation at clock ω rejects its dissipated power, in steady state, only across the boundary of its convex hull, with outward heat-flux density at most $q_{\max}$. Then*

$$P \le \pi q_{\max} \Lambda^2 = \frac{\pi q_{\max} c^2 \tau^2}{4\omega^2}. \tag{14}$$

For thermal radiation from surfaces no hotter than T_{rad} into a negligible background, $q_{\max} = \sigma T_{\mathrm{rad}}^4$. If every logically irreversible bit erasure ultimately rejects at least $k_B T_{\mathrm{rad}} \ln 2$ at the rejection temperature, then the erasure rate is at most $\pi\sigma T_{\mathrm{rad}}^3 c^2 \tau^2/(4 k_B \omega^2 \ln 2)$.

The proof combines energy balance, lemma 12, the blackbody bound on thermal emission, the Landauer cost (Landauer, 1961) and the Carnot cost of pumping heat to the radiator (appendix A.8). Reversible logic avoids the erasure term (Bennett, 1982), but not the power bound. A tenfold faster clock lowers the integrated power ceiling a hundredfold. Power P can be integrated only at clocks

$$\omega \le \omega_P = \frac{c\tau}{2\Lambda_{\min}(P)}, \qquad \Lambda_{\min}(P) = \sqrt{\frac{P}{\pi q_{\max}}}. \tag{15}$$

For the nominal solar luminosity (IAU, 2015) and $\tau = 1\,\mathrm{yr}$, $\omega_\odot = 9{,}184$ with $T_{\mathrm{rad}} = 300\,\mathrm{K}$ ($\Lambda_{\min} = 3.44\,\mathrm{au}$), and $\omega_\odot = 102{,}044$ with $1000\,\mathrm{K}$ ($0.310\,\mathrm{au}$) (figure 1b). Heat carried away by ejected matter is excluded by assumption. An engineered coolant might reach a larger $q_{\max}$, but the ω^{-2} scaling holds for any fixed $q_{\max}$.

Remark 14 (Surplus power)**.** For CITH, proposition 13 supplies a physical local ceiling for integrated computation, $\bar{B}(\omega) = P_{\max}(\omega)\Delta/b_{\mathrm{ref}}$ for a planning horizon Δ. This is the bounded-budget regime in which v8 finds a tipping budget. Growing an integrated civilization requires slowing it down, since $\omega_P \propto P^{-1/2}$. The ceiling cuts both ways. If local production is capped at $\bar{B}(\omega) < B$, its left derivative at B vanishes. v8's necessary corner condition $F'(C_L) r_j + S_j \le p_j \lambda_L = 0$ then fails in every direction with positive value. Surplus beyond the integrated capacity has no integrated use. It can be left unharvested, run by separate, non-integrated minds, or spent externally. In the last case v8's no-replication comparison of substitute bundles (Corollary 21) decides whether that spending replicates.

5.2 Clock speed against efficiency

v8 closes with an open trade-off. If energy efficiency is obtained through slower adiabatic or reversible computation, productivity a and the clock ω are coupled, and raising a can weaken the delay mechanism. In Brownian and adiabatic computing, dissipation per step grows with the speed of the step (Bennett, 1982). We model the coupling as

$$a(\omega) = \Xi\, \omega^{-\alpha}, \qquad 0 \le \alpha \le 1, \quad \omega \ge \omega_{\mathrm{lo}} > 0, \tag{16}$$

with $\alpha = 1$ for dissipation proportional to speed. For $\eta = 1$ and logarithmic production, the latency-discounted required price of a replication direction with delay T is, by v8, Corollary 6,

$$g(\omega) = e^{-\beta\omega}\Big[H + \frac{A}{\Xi}\,\omega^{\alpha}\Big], \qquad \beta = \delta T. \tag{17}$$

Proposition 15 (Clock speed and efficiency)**.** *On $[\omega_{\mathrm{lo}}, \infty)$, g is either strictly decreasing or strictly increasing and then strictly decreasing, and $g(\omega) \to 0$.*

(i) The corner set $\{\omega \ge \omega_{\mathrm{lo}} : g(\omega) \le p\}$ is $[\omega_{\mathrm{lo}}, \omega_-] \cup [\omega_+, \infty)$ with $\omega_+ < \infty$. The first interval may be empty.

(ii) If $\alpha = 0$, the corner set is $[\omega_+, \infty)$.

(iii) *If $\alpha > 0$ and $g(\omega_{\mathrm{lo}}) \le p < \max g$, the corner holds for slow and for fast clocks, but fails on the band (ω_-, ω_+).*

(iv) *As $\omega_{\mathrm{lo}} \to 0$ with $\alpha > 0$, $g(\omega_{\mathrm{lo}}) \to H$. The slow branch therefore exists asymptotically exactly when $p > H$, that is, when $B < B_{\mathrm{tip}}$.*

(v) *For $\alpha = 1$ and $\hat{\omega} = 1/\beta - H\Xi/A > 0$, the maximum is $g(\hat{\omega}) = (A/\Xi\beta)\exp(\beta H\Xi/A - 1)$.*

Exponential discounting eventually defeats any polynomial loss of efficiency, so the coupling never removes the fast route to inwardness. The efficient slow branch, however, only reproduces v8's tipping comparison, and an intermediate band of clocks is the worst place to be (figure 4a; appendix A.9). Which clock a civilization chooses depends on how it values speed. If only computation is valued and hardware is plentiful, efficiency favours slow clocks (Dyson, 1979; Sandberg et al., 2017). That is exactly where latency protection is weakest.

5.3 The light-speed containment budget

Theorem 16 (Integration-compatible latency containment)**.** *Let a civilization dissipate its entire budget B, energy per horizon Δ in units b_{ref}, in one τ-integrated computation under the flux limit of proposition 13. Let productivity a be fixed, and let every replication direction have its nearest opportunity at r_1 with delay $\kappa r_1/c$. Let $\rho(B)$ be v8's undiscounted robustness ratio over replication directions, and suppose every clock up to $\omega_{\max}(B) = c\tau/[2\Lambda_{\min}(B)]$, with $\Lambda_{\min}(B) = [Bb_{\mathrm{ref}}/(\pi q_{\max}\Delta)]^{1/2}$, is available.*

(i) *Some integration-compatible clock makes every replication direction pass its latency-discounted corner test exactly when*

$$\frac{\kappa\delta\tau}{2}\,\frac{r_1}{\Lambda_{\min}(B)} \ge \ln\rho(B). \tag{18}$$

The speed of light does not appear.

(ii) *If ρ is continuous, non-decreasing and unbounded, as in v8's baseline with a positive residual slope, the budgets satisfying equation (18) form an interval $(0, B_{\mathrm{lat}}]$ with a unique $0 < B_{\mathrm{lat}} < \infty$. Moreover $B_{\mathrm{lat}} \ge B^{(0)} = \sup\{B : \rho(B) \le 1\}$, and B_{lat} is non-decreasing in r_1, κ, $\delta\tau$ and $q_{\max}$.*

(iii) *With coupled efficiency equation (16), $\alpha \in (0, 1]$, and clocks in $[\omega_{\mathrm{lo}}, \omega_{\max}(B)]$ with $\omega_{\mathrm{lo}} \le \omega_{\max}(B)$, a containing clock exists exactly when $\min\{g(\omega_{\mathrm{lo}}), g(\omega_{\max}(B))\} \le p$. The speed of light then re-enters through $\omega_{\max}(B)^{\alpha}$.*

At the fastest admissible clock the exponent is $\delta\omega_{\max}\kappa r_1/c = (\kappa\delta\tau/2)r_1/\Lambda_{\min}(B)$. This is lemma 2 with $\Lambda = \Lambda_{\min}$ (appendix A.10). The speed of light sets the scales: how fast a civilization can think and how far away its neighbours feel. Whether an integrated civilization can be contained by latency depends instead on a geometric ratio. That ratio is the distance to the nearest opportunity divided by the smallest body able to radiate the civilization's power, multiplied by the subjective constant $\kappa\delta\tau/2$. A larger c would allow faster clocks and shorten delays in exactly offsetting proportion. If a technological speed limit below $\omega_{\max}(B)$ binds instead, it replaces $\omega_{\max}(B)$ in the test, and c re-enters.

Table 3 and figure 4(b) use the v8 one-channel coefficients with $a = 50$, $r_1 = 1.30\,\mathrm{pc}$ and $\kappa = 2$. Without latency, the corner holds up to $B^{(0)} = 96.95$ budget units, 0.80 of a solar luminosity sustained for a year. The nominal solar luminosity can be latency-contained once $\delta\tau \ge 3.3 \times 10^{-6}$ with 300 K radiators, or $\ge 3.0 \times 10^{-7}$ at 1000 K. Here $\delta\tau$ is the subjective discount rate times the subjective tolerance, both in years. Splitting the budget among N separate computations divides $\Lambda_{\min}$ by $\sqrt{N}$ and does not raise ρ. Many small, fast minds are therefore more easily contained than one large integrated mind. This is the geometric form of the inward scenario. Each small mind is, however, a separate decision-maker.

Table 3: Containment budget B_{lat} of theorem 16 for the v8 one-channel coefficients $(p, S, r, d_0, a) = (1, 0.002, 0.05, 1, 50)$, $r_1 = 1.30\,\mathrm{pc}$, $\kappa = 2$, $b_{\mathrm{ref}} = 10^{32}\,\mathrm{J}$ and a Julian-year horizon, in budget units and in nominal solar luminosity-years (120.8 units). Radiative flux limits are scenarios.

	$T_{\mathrm{rad}} = 300$ K		$T_{\mathrm{rad}} = 1000$ K	
$\delta\tau$	B_{lat}	$L_\odot$-yr	B_{lat}	$L_\odot$-yr
10^{-8}	97.02	0.803	97.74	0.809
10^{-6}	104.1	0.862	177.8	1.47
10^{-4}	1.003×10^{3}	8.30	2.347×10^{4}	194
10^{-2}	7.411×10^{5}	6.14×10^{3}	4.430×10^{7}	3.67×10^{5}
1	2.170×10^{9}	1.80×10^{7}	1.721×10^{11}	1.42×10^{9}

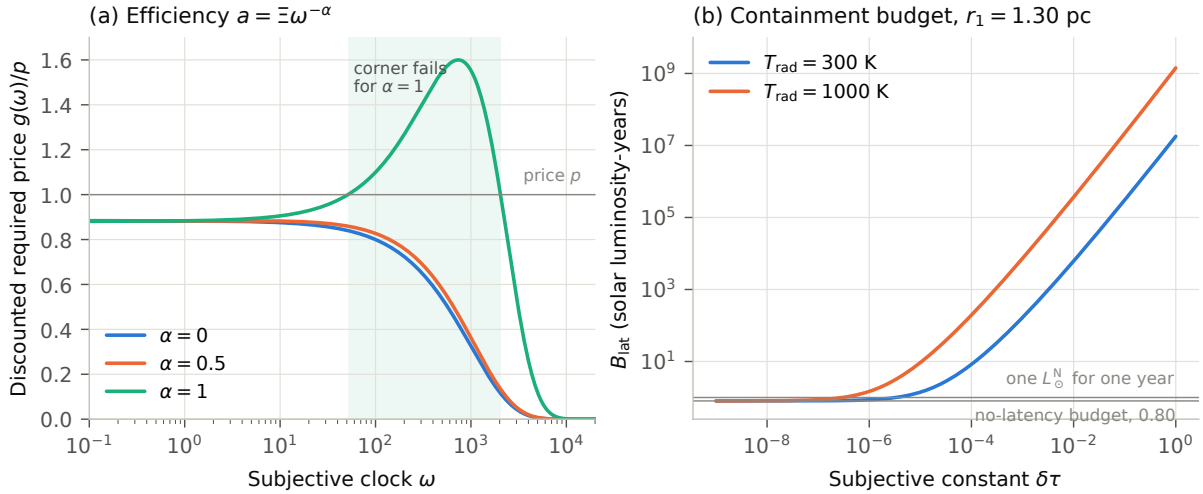


Figure 4: (a) Latency-discounted required price, relative to the actual price, under the efficiency coupling equation (16), at $B = 0.9B_{\mathrm{tip}}$ with $\beta = 10^{-3}$ and $\Xi = 1{,}500$. For $\alpha = 1$ the corner fails on $51.5 < \omega < 2{,}042$ but holds for slower and faster clocks. (b) The containment budget B_{lat} of theorem 16 against the subjective constant $\delta\tau$. All parameters are scenarios.

6 Decentralized growth: what light speed cannot contain

6.1 Colonies evaluate their own frontier

A colony beyond the coherence diameter cannot be steered from home in real time. The home's preferences reach it only through what was built in at its founding, through messages delayed by at least r/c, or through physical enforcement that is itself delayed. A colony that evaluates its own opportunities applies proposition 8 with its own nearest-neighbour distance, not with its distance from the origin. The self-limitation of proposition 9 then has no purchase, and growth becomes a question about the colony-level decision.

Proposition 17 (Lineages of autonomous colonies)**.** *Let sites be the vertices of an infinite connected graph G of maximum degree $\Delta \geq 2$, with sites one jump apart adjacent. The home is settled and founds colonies at all its neighbours. Each newly founded colony is independently an* expander *with probability π, founding colonies at all its unsettled neighbours, or founds none.*

(i) If $\pi \in \{0, 1\}$, as for identical deterministic colonies, the lineage is either the home and its neighbours or every vertex of G. A causal envelope (lemma 3) bounds the speed of the front. The colony-level test decides the outcome. The home's stopping radius does not.

(ii) If $(\Delta - 1)\pi < 1$, the lineage is finite almost surely, and

$$\mathbb{E}[\#\textit{settled}] \leq 1 + \Delta + \frac{\Delta(\Delta - 1)\pi}{1 - (\Delta - 1)\pi}, \tag{19}$$

with equality on the Δ-regular tree.

(iii) If π exceeds the site-percolation threshold $p_c^{\text{site}}(G)$, the lineage is infinite with positive probability. On the cubic lattice $p_c^{\text{site}} \simeq 0.3116$ (Deng and Blöte, 2005). On the Δ-regular tree $p_c^{\text{site}} = 1/(\Delta - 1)$.

Expanders form a branching process dominated by binomial offspring, and the settled set is an open site-percolation cluster with its boundary (appendix A.11; Grimmett, 1999). In the validator, simulated mean lineage sizes on regular trees match equation (19). On the cubic lattice the means at $\pi = 0.10$, 0.15 and 0.19 are 11.8, 17.1 and 25.7, below the tree bounds 13, 25 and 121. Lineages reached distance 30 in 94.5 per cent of runs at $\pi = 0.45$ and in none at $\pi = 0.2$.

The implication for CITH is a second uniformity requirement. v8, Proposition 27, bounds the expansionist fraction among independent origins. Proposition 17 adds a bound inside each lineage that has begun. Long jump ranges make it stringent. At 0.08 systems per pc^3, a jump range of 5 pc contains about 42 systems on average. With a maximum degree of that order, the subcritical condition of (ii) asks that fewer than about one colony in 41 become an expander. Causal separation also lets values drift independently in each colony, so restraint that relies on shared values must be inherited colony by colony. These conditions sit alongside, not in place of, the population constraint of v8.

6.2 Near-light fronts hide themselves

Remark 18 (Visibility of near-light fronts)**.** In static Euclidean space, let origins form a homogeneous Poisson process in space-time with intensity ζ, and let each origin launch a spherical front at speed $v < c$. For an observer at time t_0, consider two counts: fronts that have arrived, and origins visible but not yet arrived. They are independent Poisson variables with means $(\pi/3)\zeta v^3 t_0^4$ and $(\pi/3)\zeta(c^3 - v^3)t_0^4$. Their ratio is $(c/v)^3 - 1$. Under v8's null-arrival allowance $\mathcal{L}_* = -\ln P_*$, the expected number of visible but not yet arrived expanders is at most $[(c/v)^3 - 1]\mathcal{L}_*$. For $P_* = 0.05$ this is 1.1 at $v = 0.9c$ and 0.092 at $v = 0.99c$.

The origin event is the first part of a bubble whose light reaches the observer, which gives the two regions of the past light cone (appendix A.12). Olson (2015) and Hanson et al. (2021) develop

Table 4: What the speed of light bounds, and what follows for CITH.

Channel	Bound	Consequence for the inward scenario	Results
Acquisition	Resources are those of the light cone; polynomial growth; finite in ΛCDM	Bounded budgets; basin weakly enlarged; lock-in persists	3–6
Latency	Every remote return waits at least $\kappa r/c$	ρ becomes a distance; central expansion stops near $\ell_s \ln \ell_s$; backups are stale	7–10
Integration	Coherence diameter $c\tau/(2\omega)$	Power ceiling $\propto \omega^{-2}$; growth requires slowing; containment budget, with c cancelling	12–16
Decentralization	No real-time control beyond Λ	Lineages percolate; restraint must be inherited; near-light fronts are seen late	17, 18

cosmological versions. Light speed hides fast growth until it arrives. For near-light fronts, null visibility adds little to null arrival.

7 Discussion

Table 4 collects the four channels. The speed of light helps an inward scenario in three ways. It supplies physical reasons for bounded budgets: the cosmological ceiling, polynomial acquisition, and the coherence ceiling on integrated computation. It makes latency a consequence of geometry rather than a free parameter. And it ties fast clocks to small integrated sizes, so that growth of an integrated civilization is, in the sense of Smart (2012), a densification.

It does not fix the preference parameters on which the latency mechanism rests: the subjective discount rate, the integration tolerance and the clock. v8 already noted that subjective-time discounting is optional. The containment criterion equation (18) shows that the speed of light cannot replace such a preference, since it cancels. Light speed also leaves surplus power without an integrated use (remark 14), which can favour external projects. Above all it cannot stop autonomous descendants (proposition 17), so a latency-based explanation of silence must also explain why nearly every colony refuses its first step.

Limitations. The geometry is static and Euclidean except for lemma 6. Resource content is summarized by an envelope M that ignores clustering, stellar evolution and transport losses. Proposition 5 retains v8's myopic policy, linear residual utility and non-depreciating stock. In proposition 9 the budget envelope abstracts from the logistics of imports. The heat-flux limit and the integration tolerance are parameters, not engineering results, and lemma 12 treats only heat leaving through a convex hull. Proposition 15 and theorem 16 use v8's logarithmic baseline and a power-law coupling between efficiency and clock. In proposition 17 colony types are independent: correlated drift, coordination among colonies and active containment by the home are not modelled. Remark 18 ignores cosmological expansion and detection probabilities. The numerical work checks implementations and stated specializations. It does not test whether any civilization satisfies the assumptions.

Further work. Four extensions follow naturally. The first is the effect of light speed on idea production: the number of researchers who can collaborate in real time is bounded by the coherence region, which caps the scale effects of endogenous growth. The second is clock choice

with an explicit value for speed. The third is strategic containment, in which the home races interceptors against a deviant front. The fourth is a cosmological version of the frontier model.

8 Conclusion

The speed of light changes the question asked by CITH. In v8, physical budgets could grow without bound and remote returns arrived after a stated delay. Under causality, budgets grow at most polynomially and are finite in an accelerating universe. Remote returns are discounted on the latency length $c/(\delta\omega)$, and the region that can think as one shrinks as $1/\omega$. These facts make central expansion self-limiting, make insurance across stars stale, and give a unique containment budget for an integrated civilization. Whether such a civilization can be contained depends not on c but on the ratio of interstellar distance to its own minimal radiating size. The same facts make control of distant descendants impossible in real time. A silence explanation built on light-speed latency must therefore be hereditary: it must hold for every colony's first step, not only for the home's last.

Declarations

Data and code availability. The repository `https://github.com/wwwwwwwoooo/CITH-light-speed` contains the complete source, twelve validation scripts, their machine-readable outputs and literal logs, the figure script and the figures. The README gives the reproduction commands. The frozen v7.3 design and v8's no-replication classification are copied unchanged from the CITH repository, with their SHA-256 checksums recorded in the README.

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Use of artificial-intelligence tools. This draft was produced using Anthropic's Claude at the author's request. That includes the model extensions, proofs, numerical validators, figures and text. The implementations that generate and check results are AI-written and do not constitute independent human peer review. No AI system is an author. The human author retains responsibility for approving all claims and for the submission decision.

Ethics. No human participants, animals, personal research data or identifiable-person observational data were studied.

A Proofs

A.1 Causal envelope and exponential phases

Proof of lemma 3. Every event caused by the process lies in the causal future of the initial home ball. An event at time t and position y can lie there only if $|y - x| \leq ct$ for some

$|x| \le r_h$, so that $|y| \le r_h + ct$. A site whose resources have been influenced by time t hosts such an event, which proves the first bound. A resource delivered into the home ball by time t requires a causal chain from the ball to its site, reaching the site at some t_1, and back. Hence $|y| - r_h \le ct_1$ and $|y| - r_h \le c(t - t_1)$, so $|y| \le r_h + ct/2$. Under $M(r) \le Kr^k$, the budget is at most $B_{\text{home}} + K(r_h + ct)^k$, whose logarithm is $O(\ln t)$. This gives equation (5). □

Proof of corollary 4. The hypotheses give $B(0)e^{gt} \le B_{\text{home}} + K(r_h + ct)^k$ on $[0, T]$. Let T_1 be the maximum of the first two terms in equation (6). For $t \ge T_1$, $r_h + ct \le 2ct$ and $B_{\text{home}} \le K(2ct)^k$, so the right side is at most $2K(2ct)^k$. For $u \ge 0$, the tangent inequality $e^u \ge eu$ gives $e^y \ge (ey/m)^m$ for $y \ge 0$ and $m > 0$. With $m = 2k$ and $y = gt$,

$$B(0)\Big(\frac{eg}{2k}\Big)^{2k} t^{2k} \le 2K(2c)^k t^k,$$

so $t \le 2c(2K/B(0))^{1/k}(2k/eg)^2$. Every $t \in [T_1, T]$ obeys this bound, which proves equation (6). For v8's runaway, the bound $B_t - B_{\text{tip}} > (1 + g_X/p)^{t-t_1}(B_{t_1} - B_{\text{tip}})$ eventually exceeds any polynomial envelope, so it can hold for only finitely many periods. □

A.2 The capped basin

Proof of proposition 5. In the proof of v8, Proposition 14, the unique static optimizer $x^*(B, a)$ is non-decreasing in B and non-increasing in a.

(i) Write X_t, B_t for the uncapped path and X_t^c, B_t^c for the capped path from the same a_0. Both start at $X_0 = X_0^c = 0$. If $X_t^c \le X_t$, then $B_t^c = \bar{B} + \min\{g_X X_t^c, E_t\} \le \bar{B} + g_X X_t = B_t$, hence $x^*(B_t^c, a_t) \le x^*(B_t, a_t)$ and $X_{t+1}^c \le X_{t+1}$. Induction proves domination. For $a_0' > a_0$ in the capped model, $a_t' > a_t$. If $X_t' \le X_t$, then $B_t' \le B_t$ and $x^*(B_t', a_t') \le x^*(B_t, a_t') \le x^*(B_t, a_t)$, so induction proves monotonicity in a_0. Monotonicity in t follows because X_t and E_t are non-decreasing.

(ii) Let the uncapped path be captured, with $B_\infty = \sup_t B_t < B_{\text{tip}}$. By (i), $B_t^c \le B_\infty$ for all t. The threshold a_{crit}^ℓ is non-decreasing on $(0, B_{\text{tip}})$ (v8, Proposition 11), and $a_t \to \infty$. Choose T' with $a_t > a_{\text{crit}}^\ell(B_\infty)$ for $t \ge T'$. Then $a_t > a_{\text{crit}}^\ell(B_t^c)$ and the unique static optimizer is $x_t^c = 0$ for every $t \ge T'$. The capped budget can still rise after T' as banked stock becomes usable, but it stays at most $B_\infty < B_{\text{tip}}$. Hence the capped path is captured. The same argument, with (i)'s monotonicity in a_0, shows that a captured a_0 makes every larger initial productivity captured. By v8, Proposition 14, every $a_0 > a_{\text{sep}}$ is captured without the cap, and therefore with it.

(iii) For $t < t_E$, $B_t^c \le \bar{B} + E_t < B_{\text{tip}}$. If $\bar{B} + \sup E < B_{\text{tip}}$, the argument of (ii) applies with B_∞ replaced by $\bar{B} + \sup E$, for every $a_0 > 0$.

(iv) Capped budgets are non-decreasing, so $B_t^c \ge B_{\text{tip}}$ for all $t \ge t_1$. At or above B_{tip} no finite productivity makes the corner optimal (v8, Proposition 11). The unique optimizer is therefore positive. □

A.3 The cosmological ceiling

Proof of lemma 6. A light ray has comoving speed $\mathrm{d}\chi/\,\mathrm{d}t = c/a$, so $\chi = c\int \mathrm{d}t/a = c\int \mathrm{d}a/(a^2 H)$. A probe with peculiar speed $v < c$ has $\mathrm{d}\chi/\,\mathrm{d}t = v/a < c/a$, and travels strictly less. Since $E(a) \ge \sqrt{\Omega_\Lambda}$, the integral is at most $\int_{a_e}^\infty \mathrm{d}a/(a^2\sqrt{\Omega_\Lambda}) = 1/(a_e\sqrt{\Omega_\Lambda})$. The sites that can be influenced lie within the comoving ball of radius $\chi_{\text{EH}}(1)$, whose content is at most $(4\pi/3)\bar{n}\chi_{\text{EH}}(1)^3$. The quoted numbers are computed by adaptive quadrature and checked against the de Sitter and Einstein–de Sitter closed forms (appendix B). □

A.4 Replication certificate

Proof of lemma 7. Let (z^*, w^*, b^*) be a face optimizer with computation $C^* = C_0 + a\ell(z^*)$ and any restricted multiplier λ. The Karush–Kuhn–Tucker condition for z under a linear constraint

gives $\lambda \geq F'(C^*)a\ell'(z^*)$, with equality when $z^* > 0$. Concavity gives $\ell'(z^*) \geq \ell'(B)$, using one-sided derivatives at the endpoints, and $C^* \leq C_L$ gives $F'(C^*) \geq F'(C_L) > 0$. In direction n the derivative at the face is at most $F'(C^*)r_n + S_n$, because $R'(b^*) \leq R'(0)$ and $I'(w^*) \leq I'(0)$. Hence

$$\frac{F'(C^*)r_n + S_n}{\lambda} \leq \frac{r_n}{a\ell'(B)} + \frac{S_n}{F'(C_L)a\ell'(B)} = \frac{F'(C_L)r_n + S_n}{F'(C_L)a\ell'(B)} \leq p_n,$$

where the last inequality is equation (9). The same argument applies to direction h. Both inequalities of v8, Proposition 20 therefore hold, and that proposition makes the face optimizer global. With strict inequalities, v8's strict version excludes replication from every full optimizer. □

A.5 Latency radius

Proof of proposition 8. Necessity: if $p_j(r)\lambda_L < e^{-D_j(r)}[F'(C_L)r_j + S_j]$ for some pair, a small transfer from local investment to that direction raises utility. Sufficiency: concavity and the supporting-hyperplane bound of v8, Proposition 3, apply to any finite set of directions. Dividing by λ_L gives $p_j(r) \geq e^{-D_j(r)}P_j$. If p_j is non-decreasing and D_j is increasing in r, then $p_j(r)e^{D_j(r)}$ is non-decreasing, so the nearest distance binds. Finally, $p_j(r)e^{D_j(r)} \geq P_j$ is equivalent to $D_j(r) \geq \ln \rho_j$. With $D_j(r) = \delta\omega(T_j^0 + \kappa_j r/c)$ this gives equation (10). □

A.6 Frontier expansion

Proof of proposition 9. (i) For $0 \leq \eta \leq 1$ and $d_0 + \Omega \geq 1$, $(d_0 + \Omega)^\eta \leq d_0 + \Omega$. Hence

$$P_j \leq \frac{\bar{r} + \bar{S}(d_0 + \Omega)}{a\ell'(B)} = \Gamma(B)\Big[\frac{\bar{r} + d_0\bar{S}}{\Omega} + \bar{S}\Big] \leq K_1\Gamma(B),$$

because $\Omega = a\ell(B) \geq a_{\min}\ell(B_0) > 0$. By the curvature and budget envelopes, $K_1\Gamma(B(R)) \leq Q(R) = K_1K_\Gamma[1 + B_0 + K(1 + R)^k]^m$. Every opportunity lies at $r \geq R$, so $e^{-D_j(r)}P_j \leq e^{-\kappa R/\ell_s}Q(R)$, which is strictly below $p_{\min} \leq p_j(r)$ whenever $R > \bar{R}$. Since $\ln Q(R) = O(\ln R)$, $\bar{R}$ is finite. Strict passing in every replication direction excludes replication by lemma 7.

(ii) For $R \geq 0$, $1+B_0+K(1+R)^k \leq (1+B_0+K)(1+R)^k$. Hence $\ln(Q/p_{\min}) \leq C+km\ln(1+R)$, with $C = \max\{0, \ln(K_1K_\Gamma/p_{\min})+m\ln(1+B_0+K)\}$. Fix $\epsilon > 0$ and put $R_\epsilon = (km+\epsilon)(\ell_s/\kappa)\ln\ell_s$. Then $\kappa R_\epsilon/\ell_s - C - km\ln(1+R_\epsilon) = \epsilon\ln\ell_s - km\ln\ln\ell_s - O(1)$, which is positive for large ℓ_s. The left side is increasing for $R \geq km\ell_s/\kappa$, which includes $[R_\epsilon, \infty)$ once $\ln\ell_s \geq 1$. Thus $\bar{R} < R_\epsilon$ for large ℓ_s. For logarithmic production, $\ln(1 + B) \leq (1 + B)^{m-1}/[e(m - 1)]$ gives $\Gamma \leq K_\Gamma(1 + B)^m$ for any $m > 1$.

(iii) In the benchmark, the corner is optimal exactly when $\kappa R/\ell_s \geq \phi(R)$, where

$$\phi(R) = \ln \frac{(1 + B(R))[(r + d_0S)/a + S\ln(1 + B(R))]}{p}$$

(v8, Corollary 6, with returns discounted by e^{-D}). The objective is strictly concave, so $x^* > 0$ exactly when $\kappa R/\ell_s < \phi(R)$. Here $\phi(R) = k\ln R + O(\ln\ln R)$. The upper bound follows as in (ii) with $m = 1$. For the lower bound, put $R_- = (k-2\epsilon)(\ell_s/\kappa)\ln\ell_s$. Then $\kappa R_-/\ell_s = (k-2\epsilon)\ln\ell_s < (k-\epsilon)\ln R_- \leq \phi(R_-)$ for large ℓ_s, so expansion is optimal at R_-. Thus $R_{\text{stop}}/(\ell_s\ln\ell_s) \to k/\kappa$. For the rate, $u = \kappa R_{\text{stop}}/\ell_s$ solves $u = \phi(\ell_s u/\kappa) = k\ln\ell_s + k\ln u + O(\ln\ln(\ell_s u))$. Because $u = \Theta(\ln\ell_s)$, this gives $u = k\ln\ell_s + O(\ln\ln\ell_s)$, which is equation (12). □

A.7 Stale backups

Proof of corollary 10. The objective is $[1 - \vartheta + \vartheta\psi\varphi(x/C^\gamma)]F_\eta(C)$. This is v8, eq. (54), with φ replaced by $\psi\varphi$, which satisfies $\psi\varphi(0) = 0$, $0 \leq \psi\varphi \leq 1$ and has initial derivative $\psi\varphi'(0)$. v8's

derivation of the floor uses only these properties, so equation (13) follows. The ceiling follows as in v8 from $K_\eta \to (1-\eta)^{-1}$ and $\Gamma r/\Omega \to 0$. If valued state is proportional to e^{gs} in subjective time s, a copy lagging by $\omega r/c$ holds the fraction $e^{-g\omega r/c}$ of the current value. □

A.8 Coherence ceiling

Proof of lemma 12 and proposition 13. Taking the convex hull does not increase the diameter, and the projection of a set of diameter at most Λ has diameter at most Λ. Cauchy's formula and the planar isodiametric inequality then give a surface area of at most $4 \cdot \pi\Lambda^2/4$. A degenerate, planar hull has area $2 \cdot \pi\Lambda^2/4$. In steady state the dissipated power leaves through the hull, so $P \le q_{\max}\pi\Lambda^2$. For thermal emission by surfaces no hotter than T_{rad}, the specific intensity anywhere is at most the blackbody intensity at T_{rad}. The outward flux through any surface element is therefore at most $\sigma T_{\mathrm{rad}}^4$. An erasure at computing temperature T_c produces at least $k_B T_c \ln 2$ of heat. If $T_c < T_{\mathrm{rad}}$, a heat pump must reject at least T_{rad}/T_c times that heat at T_{rad}. If $T_c \ge T_{\mathrm{rad}}$, the heat is already at least $k_B T_{\mathrm{rad}} \ln 2$. Dividing the power bound by $k_B T_{\mathrm{rad}} \ln 2$ gives the erasure bound. □

A.9 Clock speed and efficiency

Proof of proposition 15. Write $A' = A/\Xi$. Then $g'(\omega) = e^{-\beta\omega}h(\omega)$ with $h(\omega) = \alpha A'\omega^{\alpha-1} - \beta(H + A'\omega^\alpha)$. For $\alpha \in (0,1]$ and $A' > 0$, $h'(\omega) = \alpha(\alpha-1)A'\omega^{\alpha-2} - \alpha\beta A'\omega^{\alpha-1} < 0$. For $\alpha = 0$, $h = -\beta(H + A') < 0$. The sign of g' thus changes at most once, from positive to negative. Exponential decay dominates the polynomial factor, so $g \to 0$. A unimodal or decreasing function has superlevel sets that are intervals, which gives (i). Statements (ii) and (iii) restate (i) in the decreasing and unimodal cases. For (iv), $A'\omega^\alpha \to 0$ and $e^{-\beta\omega} \to 1$ as $\omega \to 0$. The inequality $p > H = (1+B)\ln(1+B)S$ is $\Gamma(B) < p/S$, that is, $B < B_{\mathrm{tip}}$. For (v), $h(\hat\omega) = 0$ at $\hat\omega = 1/\beta - H/A'$, and substitution gives the maximum. □

A.10 Containment

Proof of theorem 16. Integration of power Bb_{ref}/Δ requires $\pi q_{\max}\Lambda^2 \ge Bb_{\mathrm{ref}}/\Delta$ by proposition 13, that is, $\omega \le \omega_{\max}(B)$. With fixed a, the discounted test $e^{-\delta\omega\kappa r_1/c}\rho(B) \le 1$ becomes easier as ω grows. A containing clock therefore exists exactly when the test passes at $\omega_{\max}(B)$, where the exponent equals $\delta\kappa r_1 c\tau/[2c\Lambda_{\min}(B)] = (\kappa\delta\tau/2)r_1/\Lambda_{\min}(B)$. This proves (i), and c has cancelled. For (ii), the left side of equation (18) is proportional to $B^{-1/2}$ and decreases strictly from $+\infty$ to 0. The right side is continuous, non-decreasing and unbounded. Their difference is strictly decreasing and changes sign exactly once. Below $B^{(0)}$ the right side is non-positive, so these budgets are feasible. The left side increases with r_1, κ, $\delta\tau$ and $q_{\max}$, so the root moves right. For (iii), proposition 15 shows that g is unimodal or decreasing on the admissible interval, so its minimum there is attained at an endpoint. Since $\omega_{\max} \propto c$, $g(\omega_{\max})$ depends on c through $\omega_{\max}^\alpha$, while $\beta\omega_{\max}$ does not. □

A.11 Lineages

Proof of proposition 17. (i) With $\pi = 0$ no colony founds another. With $\pi = 1$, induction on graph distance settles every vertex, since G is connected. The physical speed of the front is bounded by causality. (ii) Generation 0 is the home, which founds at most Δ colonies. A colony founded by an expander u has u as a settled neighbour, so it founds at most $\Delta - 1$ colonies if it expands. Types are drawn independently at founding. The expected number of expanders in generation 1 is therefore at most $\Delta\pi$, and conditional expectations contract by at most $(\Delta-1)\pi$ per generation. The expected number of non-home expanders is at most $\Delta\pi/[1-(\Delta-1)\pi]$. Settled sites number at most $1 + \Delta + (\Delta-1)$ times the non-home expanders, which proves equation (19). Finite expectation implies almost-sure finiteness. On the Δ-regular tree no

vertex can be reached twice, and every non-home vertex has exactly $\Delta - 1$ children, so every inequality is an equality. (iii) Draw all types in advance, declaring each non-home vertex open with probability π, and the home open. A vertex is settled exactly when it is the home or adjacent to a settled open vertex. The settled open vertices are therefore the open cluster of the home, and the lineage is infinite exactly when that cluster is. For $\pi > p_c^{\mathrm{site}}(G)$ this has positive probability, by the definition of the threshold and its independence of the reference vertex in a connected graph (Grimmett, 1999). On the Δ-regular tree the open cluster is a Galton–Watson tree with mean offspring $(\Delta - 1)\pi$, giving the stated threshold. □

A.12 Visibility

Proof of remark 18. Light from the near edge of a front, emitted at $s' \geq s$ by an origin at (x, s), reaches the observer at $s' + [|x| - v(s' - s)]/c = s + |x|/c + (s' - s)(1 - v/c) \geq s + |x|/c$. The origin is therefore visible by t_0 exactly when $|x| \leq c(t_0 - s)$, and its front has arrived exactly when $|x| \leq v(t_0 - s)$. The two space-time regions are disjoint, with volumes $\int_0^{t_0} (4\pi/3)v^3u^3 \, \mathrm{d}u = (\pi/3)v^3t_0^4$ and $(\pi/3)(c^3 - v^3)t_0^4$. Poisson counts on disjoint regions are independent. Conditioning on no arrival therefore leaves the visible mean unchanged. Imposing the null-arrival allowance on the arrived mean gives the bound. □

B Numerical specification

All checks use Python 3.13, NumPy 2.3.5 and SciPy 1.17.0, as pinned in `requirements.txt`. Each validator is self-contained, fixes its seed and tolerances, prints explicit pass and fail counts, and writes JSON output. No validator imports another validator's implementation. Random instances test the stated inequalities on synthetic parameters. Monte Carlo checks use margins of five standard errors. As in v8, assertion counts are not independent draws, and passing them is not evidence that any civilization satisfies the assumptions.

Table 5: Validator map. All file names are under `code/`.

Validator	Statements and implemented scope
`validate_ls_01_envelope.py`	Lemma 3 and corollary 4: random site sets and causal schedules; explicit duration bound against root finding; truncation of v8's geometric runaway.
`validate_ls_02_capped_basin.py`	Proposition 5: policy against direct maximization; domination, monotonicity, capture inclusion, crossing delay, bounded caps, lock-in; v8's separator 6.632159 and table 2.
`validate_ls_03_cosmology.py`	Lemma 6: de Sitter and Einstein–de Sitter closed forms, the $1/\sqrt{\Omega_\Lambda}$ bound, monotonicity in emission time, probes with decaying peculiar momentum, Planck numbers and radiation sensitivity.
`validate_ls_04_replication_lemma.py`	Lemma 7: exact restricted faces through the multiplier, the bound $\lambda \geq \lambda_L$, certified cases against v8's exact inequalities, sampled full optimization, and witnesses that the converse fails.
`validate_ls_05_latency_radius.py`	Proposition 8: multi-distance instances against constrained optimization; frozen-design counts under a common delay, including the exact face test; nesting and monotonicity.
`validate_ls_06_frontier.py`	Proposition 9: the general bound on random admissible instances; the benchmark's stopping radius against scalar optimization; asymptotic ratios to $\ell_s = 10^{14}$; the no-latency obstruction; figure 3.

Validator	Statements and implemented scope
`validate_ls_07_stale_backup.py`	Corollary 10: complex-step derivatives of the product objective, floor and ceiling equivalences, and an explicit staleness model.
`validate_ls_08_coherence.py`	Lemmas 2 and 12 and proposition 13: hull areas of random point clouds, Cauchy's formula by Monte Carlo projection, the planar isodiametric inequality, the Landauer–Carnot cost, and table 1.
`validate_ls_09_clock_coupling.py`	Proposition 15: unimodality, corner-set structure, agreement with direct optimization, the slow-clock limit and the adiabatic peak.
`validate_ls_10_containment.py`	Theorem 16: clock scans against equation (18), cancellation of c, uniqueness and monotonicity of B_{lat}, the endpoint test under coupling, and table 3.
`validate_ls_11_lineages.py`	Proposition 17: deterministic dichotomy, tree means against equation (19), survival probabilities, and cubic-lattice regimes.
`validate_ls_12_visibility.py`	Remark 18: Poisson space-time simulation of means, independence, ratios and conditioning on no arrival.

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